MILITARY SPECIFICATION

CONTAINER, AMMUNITION, FIBER, SPIRALLY WOUND

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for spirally wound fiber ammunition containers (See 6.1).

1.2 Classification. This specification covers four types of fiber containers and two classes of metal ends defined as follows:

Fiber Containers

Type I	-	with two layers of aluminum foil
Type II	-	with one layer of aluminum foil
Type III	-	without aluminum foil
Type IV	-	with two layers of aluminum foil and resin impregnated

Metal Ends

Class 1	-	Single crimp
Class 2	-	Double crimp

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of the specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 8140

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-16T14:56Z
Check the source to verify that this is the currentversion before use.

SPECIFICATIONS

FEDERAL

QQ-A-1876	-	Aluminum Foil
QQ-S-775	-	Steel Sheets, Carbon, Zinc Coated (Galvanized) by the Hot Dip Process
QQ-S-781	-	Strapping, Steel and Seals
QQ-T-425	-	Tin Plate (Electrolytic)
UU-C-282	-	Chipboard
UU-P-31	-	Paper: General Specification and Methods of Testing
MMM-A-100	-	Adhesive, Animal Glue
PPP-B-636	-	Box, Shipping, Fiberboard
PPP-F-320	-	Fiberboard, Corregated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-S-760	-	Strapping, Nonmetallic (and Connectors)
PPP-B-1055	-	Barrier Material, Waterproofed, Flexible

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-A-3029	-	Asphalt, Waterproofing
MIL-A-13374	-	Adhesive, Dextrin (Spiral Tube Winding for Ammunition Containers)
MIL-P-19602	-	Primer, Size Coating, Baking, for Roller Coat Application
MIL-E-19603	-	Enamel, Baking for Roller Coat Application
MIL-P-20293	-	Kraft, Asphalt - Impregnated
MIL-B-20390	-	Board, Ammunition Container
MIL-V-21064	-	Varnish, Finishing, Baking for Roller Coat Application
MIL-A-45059	-	Adhesive for Bonding Chipboard to Terneplate, Tinplate, and Zincplate

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-171	-	Finishing of Metal and Wood Surfaces

Source: https://assist.dla.mil -- Downloaded: 202640-16T14:56Z
Check the source to verify that this is the currentversion before use.

MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

(Copies of drawings, specifications, handbooks, standards and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

Interstate Commerce Commission - Uniform Freight Classification 10 - Ratings, Rules and Regulations (application for copies should be addressed to the Traffic Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606). National Motor Freight Classification - Classes and Rules (application for copies should be addressed to the National Motor Freight Traffic Association, Inc., 1616 P. Street, NW, Washington, D.C. 20036).

AMERICAN SOCIETY FOR TESTING AND MATERIALS PUBLICATIONS

ASTM B117-73	-	Salt Spray (Fog) Testing, Standard Method of
ASTM D523	-	Specular Gloss, Standard Test Method for
ASTM E376-69	-	Measuring Coating Thickness by Magnetic-Field, Recommended Practice for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specificaion, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Compliance of requirements. Requirements not shown on a drawing shall comply with the specification and the specifications referenced herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:56Z
Check the source to verify that this is the current version before use.

3.2 Outer, cover and neck tubes. Outer, cover and neck tubes shall be spirally wound. See Figure 1 (applicable to Type I, II and III containers) or Figure 2 (applicable to Type IV containers) for materials, winding sequence and other details for outer and cover tubes. See Figure 3 (applicable to Type I, II and III containers) or Figure 4 (applicable to Type IV containers) for materials, winding sequence and details for neck tubes. Windings shall be lapped as shown in Figure 5. Tube windings shall be securely adhered to each other. No waterproof layer shall be cut in subsequent crimping operations. A hairline of adhesive, not more than 1/32 inch in width or 1/64 inch in height, shall be permitted beyond the edge of the outer layer for the asphalt impregnated paper (applicable to Type I, II and III containers) or the seven point kraft (applicable to Type IV container) to which the asphalt or adhesive, as applicable, is applied on outer and cover tubes only.

3.3 Inner tube. Tubes other than those specified in 3.2 shall be either convolute or spiral wound. Unless an overlap of the windings is specified, they shall be butted at the edges. Any gap at the butt joint shall not exceed 1/16 inch. Inner tubes with greater than 1/4 inch wall thickness may be made of two or more tubes placed one within the other and securely glued together. Windings shall be glued with an adhesive complying with MMM-A-100 or MIL-A-13374. Tubes shall be glued together with an adhesive complying with MMM-A-100. Inner tubes shall be made from plain commercial chipboard, suitable for winding and shall have a clean, uniform surface. The chipboard shall meet the following minimum requirements which shall be determined by the methods specified in 4.5.7.

(a) Basis Weight - three pounds per 0.001 inch of thickness per 1,000 square feet.

(b) Bursting Strength - two points per 0.001 inch thickness.

(c) Acidity, pH - 6.5 plus or minus (±) 1.5.

3.4 Container cover. The fiber container cover shall be slide fit on the neck tube. When testing as specified in 4.5.4, the weighted cover shall slide free of the neck tube without shaking or vibration within 15 seconds.

Source: https://assist.dla.mil -- Downloaded: 202610-16T14:56Z
Check the source to verify that this is the current version before use.

3.5 Assembly. Tubes, supports and other related parts when not specifically dimensioned shall form a snug fit with mating pieces. Unless otherwise indicated on the drawing, all tubes in either the body or cover assemblies shall be securely glued together with a glue complying with MMM-A-100 or tackified polyvinyl alcohol (or acetate) adhesive. Glue shall be applied in at least one continuous band completely around the tube being glued. Tubes, supports and parts other than spacers fastened into fiber containers shall not be displaced when tested as specified in 4.5.1.

3.5.1 Tubes (except outer and cover tubes). Any inner, support or neck tube which is shown on the drawing as resting against a metal end, a support another tube, or some other part shall make contact at least at one point on the periphery of the tube. When inner support or neck tubes are required but are not illustrated on a drawing they shall contact the metal end on at least one point of the periphery of the tube. Tubes shall be cut square to their central axis. Each end shall be square within one half the tolerance permitted for the length of the tube.

3.5.2 Neck tube. Neck tubes, when glued to the outer tube, shall withstand the torque requirements specified in Table I without movement in the tube when tested as specified in 4.5.2. Any adhesive remaining on the exposed exterior surface of the neck tube after assembly shall be removed. Neck tubes shall be perforated as shown in Figure 6 to provide pressure relief in closing and opening the container. When the exposed length of the neck tube (portion holding cap) is less than 2.5 inches, vent holes will not be required.

TABLE I. Torque requirements.

Outside diameter of outside tube	Torque to be applied inch pounds minimum
3 inches or less	50
Greater than three (3) inches	100

3.6 Neck ring. Neck rings shall be applied to the neck tubes of fiber containers as required by the applicable drawing or other documents. Neck rings shall also be applied to fiber containers that open at both ends only as required on the drawing. Neck rings applied in accordance with this specification shall comply with Figure 6. The neck ring shall be flush with the outside surface of the tube and may project above the inside surface of the tube by the thickness of the metal. The ring shall be crimped in place so that no sharp corners or burrs project. The

Source: https://assist.dla.mil -- Downloaded: 202640-16T14:56Z
Check the source to verify that this is the currentversion before use.

waterproof layers of paper shall not be cut by the neck ring when it is crimped in place. All neck rings including alternate designs shall be made from 33 gage, minimum, commercial blackplate. The ring shall be blanked and plated in accordance with finish number 1.9.1.3 or 1.9.2.3 of MIL-STD-171. The completed neck rings will then be subjected to the test specified in 4.5.6.

3.6.1 Polyester tape (applicable to Type I, II and III containers. Polyester tape, instead of a neck ring, may be applied to the neck tube. The polyester tape shall be "Scotch" brand Film Tape, Black Y8459 (.0025 inch thick) "Scothpar" heat shrinkable film (a product of 3M Company, 3M Center, St. Paul, Minnesota 55101). The application of the polyester tape to the open end of the neck tube shall be accomplished by a heat shrink process (see 6.3) in accordance with Figure 7.

3.7 Metal ends. Ends for fiber containers covered by this specification shall be made of coated metal as specified below. Thickness of the metal used and the end profile shall be as specified on the container drawing or applicable document. The thickness specified shall be the thickness of the base steel or iron before plating. Completed metal ends shall be free of cracks, splits, or pinholes in the metal or the coating, including the embossed areas. The crimp of the metal end Class 1 or Class 2 to the outer tube shall be as shown on Figure 8 or 9, for the size of the fiber container being manufactured. After crimping, the metal end shall be smooth, without gathering of metal or paper at the edge of the crimp. No waterproof layer shall be cut by the crimping operation. The crimped metal ends shall withstand the load specified in Table II (applicable to Type I, II and III containers) or the applicable drawing (applicable to Type IV container). The load shall be applied parallel to the axis of the container in accordance with the test specified in 4.5.3.

TABLE II. Load requirements on crimped metal ends.

Inside diameter of outer or cover tube (CLASS 1)	Minimum removal force in pounds
1.50 to 2.39 inches	200
2.40 to 3.39 inches	300
Greater than 3.39 inches	400
Greater than 3.39 inches with 26-gage metal ends	800

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:56Z
Check the source to verify that this is the current version before use.

TABLE II. Load requirements on crimped metal ends.-Continued

Inside diameter of outer or cover tube (CLASS 2)	Minimum removal force in pounds
3.39 inches or less with 26-gage metal ends	900
3.40 to 4.70 inches with 26-gage metal ends	1200
Greater than 4.71 inches with 26-gage metal ends	1500

3.7.1 Metal for ends. The metal used shall conform to one of the materials listed in 3.7.1.1, or 3.7.1.2. More than one material may be used in a contract or purchase order provided that they are not mixed within a given lot. If a material other than that submitted in the first article lots is substituted during production, 25 coated ends fabricated from the substitute material shall be submitted to the salt spray test of 3.7.4 prior to use.

3.7.1.1 Tinplate. Tinplate shall be either electrolytic No. 100 or hot dipped Standard Cokes (or better) in accordance with QQ-T-425. Steel type, temper, finish, and base weight shall be selected by the fiber container manufacturer to the values which will permit compliance with 3.7.

3.7.1.2 Zinc coated steel. Zinc coated steel shall comply with Type I, Class E of QQ-S-775.

3.7.2 Metal end marking. Metal ends shall be legibly embossed as required on the applicable fiber container drawing. In addition, the manufacturer's symbol and month and year of manufacture shall be included. The month shall be represented by one or two digits, as applicable, and the year shall be represented by the last two digit of the year; a hyphen shall separate month and year. The letter "F" shall be embossed on containers when one layer of foil is included in the tube construction. "FF" shall be included when two layers of foil are included in the construction and "FFR" shall be embossed when two layers of foil are included in the resin impregnation tube construction. If necessary, the word "CONTAINER" may be abbreviated to "CNTNR" and the embossing design shown on the drawing may be shifted circumferentially to provide additional space in the embossing circle. Alternatively, the date and the manufacturer's symbol may be placed in the center of the embossed end.

Source: https://assist.dla.mil -- Downloaded: 202610-16T14:56Z
Check the source to verify that this is the current version before use.

3.7.3 Coating. Metal ends shall be coated on the outside surface in accordance with MIL-P-19602 and Class 2 of MIL-E-19603. A finishing varnish complying with Class 2 of MIL-V-21064 may be applied if desired. Alternately, Class 1 instead of Class 2 MIL-E-19603 may be used if finishing varnish, Class 2 of MIL-V-21064 is used. Gloss requirements of the coated end must not exceed 50 when tested as specified in 4.5.5.1. Alternately, the outside surface may be coated with finish number 20.5 of MIL-STD-171. Either coating shall be continuous and uniform. The color of the coating shall be black.

3.7.4 Salt spray. The coated ends shall be free of rust, undercutting, blistering and other defects to the extent specified in the applicable enamel specifications when tested as specified in 4.5.5.2. First article samples shall be tested for 168 hours. During production each lot of enamel coated metal sheets, as delivered to the fiber container manufacturer, shall meet the 168 hours salt spray test.

3.8 Chipboard spacers. Chipboard spacers, as required by the drawing or other document, shall be securely adhered to the inner surface of flush panel metal ends using an adhesive that complies with MIL-A-45059. Spacers shall be built up to the required thickness from chipboard complying with UU-C-282 using adhesives complying with MIL-A-13374 or MMM-A-100. The spacer shall not dislodge when the container is tested in accordance with 4.5.1. Dislodgement of the spacer caused by separation of chipboard fibers rather than by separation of the adhesive bond between the spacer and the metal end shall not be cause for rejection (see 6.4).

3.9 Peel strength of polyester tape from the surface of the outer kraft ply of the neck tube (applicable to Type I, II and III containers). The outer kraft ply of the neck tube shall exhibit 75 percent minimum (min.) fiber failure when the polyester tape is peeled when tested as specified in 4.5.8. Fiber failure does not apply to the overlap position of the tape.

3.10 Peel strength of outer layer of asphalt impregnated kraft paper to aluminum foil (applicable to Type I and II containers). The fiber container (outer and cover tube) shall exhibit 75 percent min. fiber failure when tested as specified in 4.5.9.

3.11 Resin content (applicable to Type IV containers). The resin content of the outer, cover and neck tubes shall be 12 percent min., by weight, when immersed in resin (see 6.12). Outer tubes, cover tubes, neck tubes, support tubes and spacer discs shall be separately immersed (cooked) in heated paraffin in accordance with 4.5.10. Resin content of inner support tubes and other components will not be required by this specification. However, certification, of resin impregnation, by the contractor will be required for these components.

Source: https://assist.dla.mil -- Downloaded: 202610-16T14:56Z
Check the source to verify that this is the current version before use.

3.12 First article inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.13 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be clean and free from surface scuffing, imperfect adhesion between layers, bleeding of asphalt (applicable to Type I, II and III containers), blisters (see 6.7), checking of paper on inside and outside surfaces, ragged slitter and recutter cuts, burrs on the inside edge of the open end of the cover, ragged edges on the open end of other tubes, dirt, grease, rust, and other foreign matter except that traces of process lubricant will be permitted provided that noticeable staining or discoloration does not occur. If wax is used as a lubricant, transfer of the wax to the inner ply of the container shall be permitted but there shall be no evidence on the inner plies of loose particles, flaking or caking of the wax (see 6.5).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define term used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:56Z
Check the source to verify that this is the current version before use.

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of twenty five (25) completed assemblies, ten (10) each of every component and every subassembly, which have been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. The contractor shall submit his test results for the resin content to show compliance with the applicable requirement. All material shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspections to be performed. Samples may be subjected by the Government to any or all of the examinations and tests (see Table III) specified in this specification and to any or all requirements of the applicable drawings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:56Z
Check the source to verify that this is the current version before use.

TABLE III. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-C-2439E

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.3.2.1	Assemblies and Components				Applicable Dwg.
					NEXT HIGHER ASSEMBLY
					NA
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Container assemblies				
	a. Examination for defects (see Note 1)	10		3.1	4.4.2.1
	b. Drop test	5		3.5	4.5.1
	c. Torque test	5		3.5.2	4.5.2
	d. Metal end pull test	5		3.7	4.5.3
	e. Fit of cover assembly	5		3.4	4.5.4
	f. Peel strength of outer layer of asphalt impregnated kraft paper to aluminum foil (applicable to Type I and Type II containers)	5		3.10	4.5.9
	Coated metal ends				
	a. Gloss	5		3.7.3	4.5.5.1
	Coated metal ends (or materials (sheets) used for manufacture of metal ends)				
	a. Salt spray test	5		3.7.4	4.5.5.2
	Neck rings				
	a. Salt spray test	5		3.6	4.5.6.1
	b. Plating thickness	5		3.6	4.5.6.2
	Inner tubes				
	a. Basis weight	3		3.1	4.5.7.1
	b. Bursting strength	3		3.1	4.5.7.2
	c. Acidity, pH	4		3.1	4.5.7.3

NOTES: 1. The samples which are used for the Examination for Defects shall then be used for the other tests.

Source: https://assist.dla.mil -- Downloaded: 2026-01-16T14:56Z
Check the source to verify that this is the current version before use.

TABLE III. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-C-2439E

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER Applicable Dwg.
4.3.2.1	Assemblies and Components				NEXT HIGHER ASSEMBLY NA
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Neck tubes				
	a. Peel strength of polyester tape (applicable to Type I, II and III containers)	5		3.9	4.5.8
	b. Resin content (applicable to Type IV containers) (See Note 2)	5		3.11	4.5.10
	Cover tubes				
	a. Resin content (applicable to Type IV containers) (See Note 2)	5		3.11	4.5.10
	Outer tubes				
	a. Resin content (applicable to Type IV containers) (See Note 2)	5		3.11	4.5.10

NOTES: 2. For the resin content test, five labelled samples (neck, cover and outer tubes) which have not been resin impregnated shall be used.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:56Z
Check the source to verify that this is the current version before use.

4.3.3 Rejection. If any assembly or component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly or component to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submissions of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examination. Unless otherwise specified in the Classification of Defects and test tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105. Inspection Level II, except that continuous plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:56Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 3	DRAWING NUMBER See applicable drawing
4.4.2.1	Container Assembly		NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Metal end diameter		0.40%	3.1	SMTE (see 6.6)
102.	Distance from outer tube to end of neck tube (when applicable)		0.40%	3.1	SMTE
103.	Hole or break in end metal		0.40%	3.1	Visual
104.	Crimp faulty or incomplete		0.40%	3.1	Visual
105.	Cover and outer tube fail to meet		0.40%	3.1	Visual
106.	Components missing or incorrectly positioned or loose (when tilted)		0.40%	3.1	Visual
107.	Cuts, scuffs, gouges, checks, or crimping cuts which penetrate outer layer		0.40%	3.1	Visual
Minor					
201.	Length of outer tube		0.65%	3.1	SMTE
202.	Length of outer cover		0.65%	3.1	SMTE
203.	Depth to support or support assembly from top of neck tube (or neck ring)		0.65%	3.1	SMTE
204.	Depth to spacer tube from top of neck tube (or neck ring)		0.65%	3.1	SMTE

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:56Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER
4.4.2.1	Container Assembly				See applicable drawing
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
205.	Outside diameter of outer tube or cover tube minimum (min.)		0.65%	3.1	SMTF
206.	Inside diameter or neck tube (or neck ring)		0.65%	3.1	SMTF
207.	Inside diameter of support tube (as applicable)		0.65%	3.1	SMTF
208.	Inside diameter of spacer tube		0.65%	3.1	SMTF
209.	Embossing on end missing, misleading or unidentifiable		0.65%	3.1	Visual
210.	Glue (dry or tacky) on outside of neck or inside diameter of cap which could interfere with closing or at any point of the interior which will be in contact with the intended contents of container		0.65%	3.1	Visual
211.	Looseness between layers		0.65%	3.1	Visual/Manual
212.	Support or support assembly damaged or deformed		0.65%	3.1	Visual
213.	Neck ring or support ring improperly crimped		0.65%	3.1	Visual
214.	Metal end crimped to wrong end of tube		0.65%	3.1	Visual
215.	Cuts, scuffs, gouges or checks not penetrating outer layers		0.65%	3.1	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-04-26 16:52Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 3 OF 3	DRAWING NUMBER See applicable drawing
4.4.2.1	Container Assembly		NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
216.	Marking missing, misleading or unidentifiable		0.65%	3.1	Visual
217.	Mildew or mold present		0.65%	3.1	Visual
218.	Staple mislocated, missing, broken or improperly clinched (as applicable)		0.65%	3.1	Visual
219.	Bleeding of asphalt on outer surface (applicable to Type I, II and III containers)		0.65%	3.13	Visual
220.	Blisters (see 6.7) (applicable to Type I, II and III containers)		0.65%	3.13	4.5.11/Visual
221.	Evidence of poor workmanship		0.65%	3.13	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:56Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-2439E

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER: See applicable drawing
4.4.2.2	Shipping Container, (Box) Sealed				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Container improperly sealed		0.40%	3.1	Visual
102.	Contents loose (when tilted)		0.40%	3.1	Manual
103.	Container cut or torn		0.40%	3.1	Visual
Minor					
201.	Marking missing, incorrect or unidentifiable		0.65%	3.1	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-0-16 14:56Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-2439E

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Strapped Bundles	1 OF 1	See applicable drawing
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Strapping loose		0 40%	3.1	Visual/Manual
102.	Chipboard missing (see 5.2.2.2.1)		0.40%	3.1	Visual
Minor					
201.	Marking missing, misleading or unidentifiable		0.65%	3.1	Visual

NOTES:

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:56Z
Check the source to verify that this is the current version before use.

4.4.3 Testing.

4.4.3.1 Container with displaced parts. (see 3.5 and 3.8), Major defect - Containers shall be randomly selected from each lot in accordance with MIL-STD-105, Inspection Level S-4, using an AQL of 2.5 percent and shall be tested for compliance with 4.5.1.

4.4.3.2 Container with less torque than specified. (see 3.5.2), Major defect - Containers shall be randomly selected from each lot in accordance with MIL-STD-105, Inspection Level S-4, using an AQL of 2.5 percent and shall be tested for compliance with 4.5.2.

4.4.3.3 Metal end stripped from container. (see 3.7), Major defect - Containers shall be randomly selected from each lot in accordance with MIL-STD-105, Inspection Level S-4, using an AQL of 2.5 percent and shall be tested for compliance with 4.5.3.

4.4.3.4 Container with loose cover. (see 3.4), Major defect - Container shall be randomly selected from each lot in accordance with MIL-STD-105, Inspection Level S-4, using an AQL of 2.5 percent and shall be tested for compliance with 4.5.4.

4.4.3.5 Coated metal ends.

4.4.3.5.1 Gloss of coated metal ends. (see 3.7.3), Major defect - Coated metal ends shall be randomly selected from each lot in accordance with MIL-STD-105, Inspection Level S-4, using an AQL of 4.0 percent and tested for compliance with 4.5.5.1.

4.5.3.5.2 Salt spray test of coated metal ends. (see 3.7.4), Major defect - Sampling of each lot of metal ends (or material lots used in the manufacture thereof) shall be in accordance with the applicable enamel specifications and tested for compliance with 4.5.5.2.

4.4.3.6 Neck rings.

4.4.3.6.1 Salt spray test. (see 3.6), Major defect - Fifteen (15) components shall be randomly selected from each lot and subjected to this test. Failure of one or more neck ring to meet the applicable requirement, when tested as specified in 4.5.6.1, shall be cause for rejection of the lot.

4.4.3.6.2 Plating thickness. (see 3.6), Major defect - Fifteen (15) components shall be randomly selected from each lot and subjected to this test. Failure of one or more neck ring to meet the applicable requirements, when tested as specified in 4.5.6.2, shall be cause for rejection of the lot.

Source: https://assist.dla.mil -- Downloaded: 202610-16T14:56Z
Check the source to verify that this is the current version before use.

4.4.3.7 Inner tubes.

4.4.3.7.1 Basis weight. (see 3.3), Major defect - Inner tubes shall be randomly selected from each lot in accordance with MIL-STD-105, Inspection Level S-4, usin an AQL of 2.5 percent and shall be tested for compliance with 4.5.7.1.

4.4.3.7.2 Bursting strength. (see 3.3), Major defect - Inner tubes shall be randomly selected for each lot in accordance with MIL-STD-105, Inspection Level S-4, using an AQL of 2.5 percent and shall be tested for compliance with 4.5.7.2.

4.4.3.7.3 Acidity, pH. (see 3.3), Major defect - Inner tubes shall be randomly selected from each lot in accordance with MIL-STD-105, Inspection Level S-4, using an AQL of 2.5 percent and shall be tested for compliance with 4.5.7.3.

4.4.3.8 Peel strength of polyester tape from the surface of the outer kraft ply of neck tube (applicable to Type I, II and III containers). see (3.9), Major defect - Fifteen (15) container assemblies shall be selected from each lot for this test. If two or more assemblies fail to meet the applicable requirement when tested as specified in 4.5.8, the lot shall be rejected.

4.4.3.9 Peel strength of outer layer of asphalt impregnated kraft paper to aluminum foil (applicable to Type I, II and III container). (see 3.10), Major defect - Five (5) container assemblies shall be selected from each lot for this test. From each container assembly selected, two (2) test coupons (measuring 1 inch min. wide by four inch min. long) shall be prepared and subjected to the test specified in 4.5.9. If one or two coupons fail to meet the applicable requirement, the lot shall be eligible for retest as specified in 4.4.3.9.1. Failure of three or more coupons shall be cause for rejection of the lot.

4.4.3.9.1 Retest. If one or two test coupons fail to comply with the applicable requirement, the lot shall be considered eligible for retest if requested by the contractor. A sample of fifteen (15) additional assemblies shall be selected and the coupons prepared as specified in 4.4.3.9. Failure of a cumulative total of three coupons to comply with the applicable requirement, when tested as specified in 4.5.9, shall be cause for rejection of the lot.

Source: https://assist.dla.mil -- Downloaded: 202610-16T14:56Z
Check the source to verify that this is the current version before use.

4.4.4 Inspection equipment. The government reserves the right to inspect the contractor's equipment and determine that he has available and utilizes correctly, measuring and test equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. Commercial inspection equipment shall be employed where applicable for all tests and examinations specified in 4.4.2 and 4.5. The contractor is responsible for assuring that proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purposes (see 6.8).

4.5 Methods of inspection. (see 6.9)

4.5.1 Drop test. The container shall be held vertically with the low end four (4) feet from a concrete or steel surface, released and allowed to fall freely. Repeat the operation having the opposite end contact the surface. Observation shall be made for loose or misplaced parts (see 3.5).

4.5.2 Torque test. The container shall be assembled into the fixture shown on Figure 10, or an approved equivalent thereto, and the required torque applied (see Table I). Care shall be exercised to prevent damage to the crimp of the metal end of the container tube.

4.5.3 Metal end pull test. This test shall be performed on the metal ends of the cover and the body assembly. The assembly shall be prepared for the test by cutting the body tube 5 to 6 inchs from the end and extracting all component parts, where necessary, from the cover or truncated body assembly without injury to the crimp of the metal end of the tube. The prepared cover and body assemblies shall then be assembled in an approved fixture similar to that shown in Figure 11. The test may be performed in a tensile testing machine or in a static testing arrangement satisfactory to the Government. Determine the load required to separate the metal end from the tube by applying the load parallel to the centerline of the specimen. When a tensile testing machine is used, the rate of travel of the head shall be 0.25 $\pm$ 0.05 inch per minute.

4.5.4 Fit of cover assembly. The container shall be placed in a vertical position, cover end down, and supported at the base. The cover shall be positioned such that the open end of the cover is just below the neck perforations. The cover shall then be released. If the cover remains in place when released, it shall be removed, a 3 pound weight shall be placed therein and then be replaced on the neck tube as far as the perforations. Disengagement of the cover from the neck shall take place within 15 seconds.

Source: https://assist.dla.mil -- Downloaded: 202640-16T14:56Z
Check the source to verify that this is the currentversion before use.

4.5.5 Coated metal ends.

4.5.5.1 Gloss of coated metal ends. The coated metal ends shall be tested in accordance with ASTM-D523.

4.5.5.2 Salt spray of coated metal ends. The coated metal ends shall be tested as specified in the applicable enamel specification except that the specimens need not be bent.

4.5.6 Neck rings.

4.5.6.1 Salt spray test. The neck rings shall be tested as specified in ASTM B117-73. The test period for the salt spray test for the neck ring shall be thirty six (36) hours.

4.5.6.2 Plating thickness. The neck rings shall be tested in accordance with ASTM E376-69 (Part 31).

4.5.7 Inner tube.

4.5.7.1 Basis weight. The inner tubes shall be tested in accordance with Method No. 110 of UU-P-31.

4.5.7.2 Bursting strength. The inner tubes shall be tested in accordance with Method No. 112 of UU-P-31.

4.5.7.3 Acidity, pH. The inner tubes shall be tested in accordance with Method No. 200 of UU-P-31.

4.5.8 Peel strength of polyester tape from the surface of the outer kraft ply of the neck tube (applicable to Type I, II and III containers). Store container assemblies without cover tubes, at ambient temperature and humidity for three (3) hours min. Upon completion of the conditioning, the tape shall be separated by a sharp object or other suitable means from the surface of the neck tube. The separation will be sufficient in length to permit fingertip grasping of the tape. Peel the tape from the neck tube at 90 degrees (^{o}) at an even rate of peel of 1 inch approximately, every five seconds. Examine for evidence of fiber failure of the surface of the outer kraft ply to the tape.

4.5.9 Peel strength of outer layer of asphalt impregnated kraft paper to aluminum foil (applicable to Type I and II containers). Prior to being subjected to this test, the test coupons shall be conditioned to the following requirement: Storage at ambient room temperature and humidity for a period of three (3) hours min. Upon completion of conditioning, the outer layer of asphalt impregnated kraft paper and adjacent layer of aluminum foil shall be separated by a sharp object or other suitable means. The separation will be sufficient in length to

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:56Z
Check the source to verify that this is the current version before use.

permit fingertip grasping of the layers. The separated layer will be peeled at a 180 degree angle and at even rate of peel of 1 inch approximately every five seconds. Examine for evidence of fiber failure of the surface of asphalt impregnated kraft paper to the layer of aluminum foil.

4.5.10 Resin content (process control of resin impregnated procedure) (applicable to Type IV containers).

4.5.10.1 Batch process. In place of the normal sampling associated with the Classification of Defects and after a cure time for the parts (outer, cover or neck tubes) has been approved (see 4.5.10.3), five (5) samples shall be inspected to qualify the resin impregnation procedure at the beginning of production. The non-impregnated samples shall be tared and labelled. Then they shall be immersed in heated resin at 200°F to 225°F for three (3) minutes (min). After the samples have been cured, they shall be weighed again and the resin content shall be calculated as follows:

$$\text{Resin content, \%} = \frac{(W2-W1)}{W2} \times 100$$

Where: W1 = Weight (in grams) before resin impregnation

W2 = Weight (in grams) after resin impregnation

If any sample fails to meet the applicable requirement, the impregnation procedure shall be adjusted and the test repeated. During production, after every four (4) hours and during the last hour of production (if applicable), two samples shall be inspected as described above. If any sample fails to meet the required resin content, that portion of production since the last control check shall be removed from the lot. The contractor may request a change of inspection frequency, provided he presents objective evidence to the contracting officer, to substantiate the request.

4.5.10.2 Continuous process. In place of the normal sampling associated with the Classification of Defects and after a cure time for the parts (outer, cover or neck tubes) has been approved (see 4.5.10.3), five (5) samples shall be inspected to qualify the resin procedure at the beginning of production as described in 4.5.10.1 During production, after every two hours and the last hour of production (if applicable), two tared and labelled samples shall be removed from the line and inspected as described in 4.5.10.1. If any sample fails to meet the required resin content, that portion of production since the last control check shall be removed from the lot. The contractor may request a change of inspection frequency provided he presents objective evidence to the contracting officer to substantiate the request.

Source: https://assist.dla.mil -- Downloaded: 202640-16T14:56Z
Check the source to verify that this is the current version before use.

4.5.10.3 Cure time. In establishing a cure time, check weigh ten (10) parts (outer, cover or neck tubes) after resin impregnation at periodic intervals (e.g. every 10, 20, 30 minutes etc.) since resin impregnation and continue until weighing stability is obtained. The cure time will be established from the time resin impregnation of the part is completed to the time weighing stability is obtained. The inspection data, used in determining the cure time, shall be sent to Commander, U.S. Army Armament Research, Development and Engineering Center, AMSMC-QAR-Q (D), Dover, New Jersey 07801-5001.

4.5.11 Blister detection (applicable to Type I, II and III containers). The fiber container with cover removed, shall be dried in a circulating oven for a period of 8 hours ± 1/2 hours at a temperature of 125 degrees (°) Fahrenheit (F) ± 5°F. Following this procedure, visual determination shall be made for indications of blistering.

5. PACKAGING

5.1 Packaging requirements. None required.

5.2 Packing.

5.2.1 Level A. Fiber containers shall be packed in boxes complying with V3c or V3s, Type RSC, CSSC, OSC, FOL or SFF of PPP-B-636. The length of the fiber containers shall be parallel to the depth of the box. When fiber containers are arranged in more than one layer in depth (of the box), pads of V3c, V3s, W5c or W5s fiberboard complying with PPP-F-320 shall be inserted between layers. Closure of the boxes shall be in accordance with the box specification.

5.2.2 Level C. Unless otherwise specified, fiber containers shall be packed in boxes or bundled as specified below.

5.2.2.1 Boxes. Fiber containers shall be packed in boxes complying with grade V3c and V3s of PPP-B-636. Type and arrangement shall be as specified in 5.2.1. Inter-layer pads shall be made from domestic fiberboard or chipboard of sufficient thickness to prevent abrasion of ends during shipment and handling. Boxes shall be closed in accordance with the box specification.

5.2.2.2 Bundles. Fiber containers 18 inches or more in length may be bundled using steel strapping. Containers more than 3 inches in diameter shall be bundled 6 to a triangular bundle or 7 to 19 to a hexagonal bundle (see Figure 12).

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:56Z
Check the source to verify that this is the current version before use.

5.2.2.2.1 Strapping. Bundles shall be strapped with steel strapping not less than 3/8 inch wide by .015 thick, Type I, Class A or C complying with QQ-S-781. Alternatively nonmetallic strapping of minimum nominal width of 3/8 inches by minimum nominal thickness of .015 Type II complying with PPP-S-760 may be used. Two straps shall be used per bundle. Chipboard not less than .025 thick, shall be used between the steel strapping and the container.

5.3 Marking.

5.3.1 Level A. The containers of Leval A shipments shall be marked in accordance with MIL-STD-1169 or as directed by the procuring activity.

5.3.2 Level C. The containers or bundles of Level C shipments shall be marked in accordance with MIL-STD-1169.

5.4 Loading of railroad cars, trucks, or trailers.

5.4.1 Preparation for bundle loading. Vehicles shall be examined to assure that roofs, floors, and walls are sound. Vehicles shall be cleaned, with particular care taken to remove protruding nails or fasteners. Half-lined rail cars may be used provided temporary sheathing, not less than 6 inches wide, is nailed to the walls at heights which coincide with the junction of the bundle layers. A liner material of .030 inch minimum thickness chipboard or fiberboard either corrugated or solid of equivalent protective value shall be used to line the vehicle walls and floor, and to cover projecting doorways, parts or projecting edges.

5.4.2 Stowing. Fiber containers, boxed or bundled shall be loaded into the railroad car, truck or trailer in a manner that affords adequate protection against physical damage during shipment and will insure safe delivery of containers from the manufacturer to the first receiving activity. Packing of the container shall conform to requirements of Interstate Commerce Commission Regulations for Uniform Freight Classification and National Motor Freight Classification in effect at time of shipment.

6. NOTES

6.1 Intended use. This specification is intended primarily for use in connection with assembly drawings of fiber containers which are intended for packing of loaded ammunition items. Details, supports, specific tube dimensions and requirements for neck ring will be as shown on the assembly drawings. Dimensions and specific details are to be determined by the design agency.

Source: https://assist.dla.mil -- Downloaded: 202610-16T14:56Z
Check the source to verify that this is the currentversion before use.

6.2 Ordering data. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.

b. Drawing number of container.

c. Type and class of container.

d. Provisions for submission of first article samples.

6.2.2 Contract data requirements. Contract data requirements for inspection equipment designs (conforming to Data Item Description DI-R-1714 tailored). See 6.8.

6.3 Suggested source. A suggested source for obtaining tape applying and heat shrink equipment is 3M Company, 3M Center, Tape Customer Engineering, St. Paul, Minnesota 55101.

6.4 Spacer material. Alternate material for chipboard is Coresote Industrial Board, Product Homosote Co., Trenton, New Jersey.

6.5 Aid in manufacturing process. As an aid in the manufacturing process, paraffin wax meeting the requirements of VV-W-95, Type I, may be used to facilitate release of tubes from the forming mandrel. If it is desired to use lubricant other than wax, they shall be approved by the procuring agency prior to use.

6.6 Standard measuring and test equipment. Where Standard Measuring and Test Equipment (SMTE) is specified as the method of inspection, the contractor may use any type of industry-developed, commercially available, multi-usage equipment or special inspection or testing equipment approved by the Government. The contractor shall be responsible for utilizing SMTE which will assure full form for the entire surface of the specified characteristic.

6.7 Blisters. It is generally believed that blistering is caused by faulty winding techniques and skips in the application of adhesive between the outer layers of the fiber container. Blistering is not always immediately and readily apparent upon completion of the winding operation. Blistering may appear after the containers have had a chance to become conditioned to ambient conditions. Exposure to large changes in temperature and humidity also tend to aggravate the condition. If there is a recent history of blistering or it is suspected that they may occur, it is recommended that the containers be held at least 24 hours and preferably 48 hours prior to final inspection for this defect.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:56Z
Check the source to verify that this is the current version before use.

6.8 Inspection equipment design. Design responsibility for inspection equipment is assigned to the contractor.

6.8.1 Contractor designs. Contractor designs are required for all inspection equipment and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment. Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The item detail specification number, paragraph number, and defect number from Section 4 shall be referenced on each contractor design together with the component or assembly drawing number, revision letter and date to which the specific design applies.

6.8.2 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contact. The specific segment of the U.S. Army Armament Research, Development and Engineering Center to which the Contractor designs shall be sent will be specified in the item detail specification. When the contractor submits inspection equipment designs to the Government for approval he shall give the following information in his letter of transmittal:

a. The contract number.

b. The contract item (name, model number, etc.).

c. The designs remaining to be submitted and the expected date of submittal.

Submit equipment designs, as required, to Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAR-I(D), Dover, New Jersey 07801-5001.

Source: https://assist.dla.mil --, Downloaded: 2026-10-16T14:56Z
Check the source to verify that this is the current version before use.

6.9 Equivalent test methods. Prior approval of the contracting officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAR-Q(D), Dover, New Jersey 07801-5001. This description should include but not be limited to the accuracy and precision of the method, test data demonstrating the accuracy and precision and drawings of any special equipment required.

6.10 International standardization agreement. Provisions for this specification are the subject of international standardization agreement. When amendment, revision or cancellation of this specification is proposed, the departmental custodians will inform their respective departmental standardization offices so that appropriate action may be taken respecting the international agreement concerned.

6.11 Packing directly in preservative treated boxes. When Type II or III containers are packed directly in preservative treated boxes, bleeding of the asphalt can be expected due to the action of solvent vapors from the preservative treatment unless a water base preservative is used.

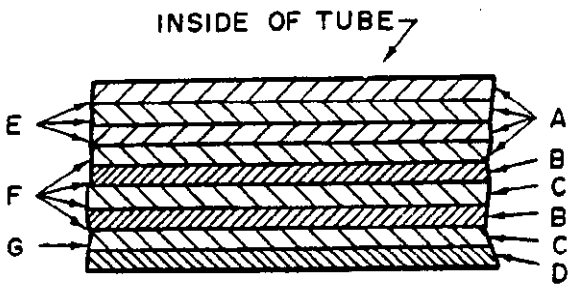
6.12 Resin material. Paraffin 3420, manufactured by Sun Oil Company, Marcus Hook, PA has been found satisfactory for use in resin immersion containers.

6.13 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

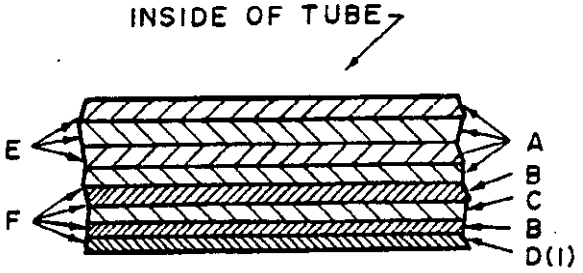
Source: https://assist.dla.mil -- Downloaded: 20261O-16T14:56Z
Check the source to verify that this is the current version before use.

CONSTRUCTION OF OUTER AND COVER TUBES

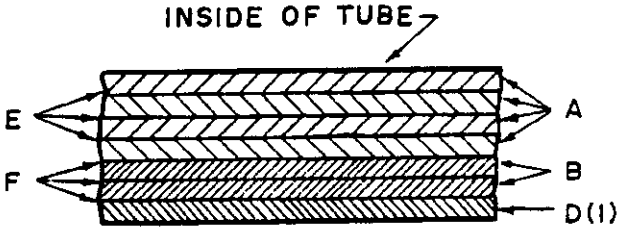
TYPE I



TYPE II



TYPE III



A - AMMUNITION CONTAINER BOARD, SPECIFICATION MIL-B-20030, AS REQUIRED TO PROVIDE SPECIFIED DIAMETER. (SEE FIGURE 1A, NOTE A)

B - .007" WATERPROOF ASPHALT DUPLEX KRAFT WRAPPING PAPER, PPP-B-1055, Class P-1 (MUST LAP NOT LESS THAN 1/16"). INNER LAYER TO BE APPLIED WITH ASPHALT ON BOTH SIDES, OUTER COVERING TO BE IMMERSED IN ASPHALT.

C - ALUMINUM FOIL, SPECIFICATION QQ-A-1876, 0.001" THICK. FOIL MAY BE PARTIALLY ANNEALED TEMPER WITH MAXIMUM BURSTING STRENGTH OF 45 PSI. LAP OF FOIL TO BE NOT LESS THAN 1/4".

D - 0.006" TO 0.007" ASPHALT IMPREGNATED KRAFT PAPER, SPECIFICATION MIL-P-20293. (LAP NOT LESS THAN 5/16").

E - GLUE, SPECIFICATION MMM-A-100. (SEE NOTE H)

F - ASPHALT, SPECIFICATION MIL-A-3029.

G - NEOPRENE LATEX BASE ADHESIVE OR POLYVINYL ACETATE ADHESIVE MEETING REQUIREMENTS OF SPECIFICATION MIL-A-45059

H - ALTERNATE ADHESIVES - ADHESIVE, DEXTRIN - SPECIFICATION MIL-A-13374 (ORD), TACKIFIED POLYVINYL ALCOHOL ADHESIVE OR POLYVINYL ACETATE ADHESIVE MEETING PERFORMANCE TESTS OF SPECIFICATION MMM-A-100.

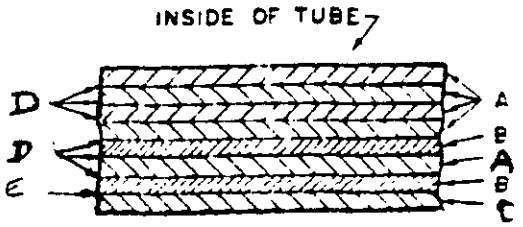
(1) ENTIRE INNER SURFACE TO BE COMPLETELY COATED WITH ASPHALT.

FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 202610-16T14:56Z
Check the source to verify that this is the currentversion before use.

CONSTRUCTION OF OUTER AND COVER TUBES

TYPE IV



A - Ammunition container board, specification MIL-B-20390, as required to provide specified diameter. Use one or more layers of paper of equal or better grade with minimum thickness of .007" to obtain specified diameter.

B - Aluminum foil, specification QQ-A-1876, 0.001" thick. Foil may be partially annealed temper with maximum bursting strength of 45 psi. Lap of foil to be not less than ½.

C - 7 point lusterless black kraft. Lap shall not be less than 5/16".

D - Glue specification MMM-A-100 (see note F)

E - Polyvinyl acetate adhesive, product no. 53-1430, supplied by United Resin Products Inc., Brooklyn, NY or equivalent.

F - Alternative Adhesive - neoprene latex base adhesive or polyvinyl acetate adhesive meeting requirements of specification MIL-A-45059.

G - Resin is applied to all the outer surfaces of the tubes.

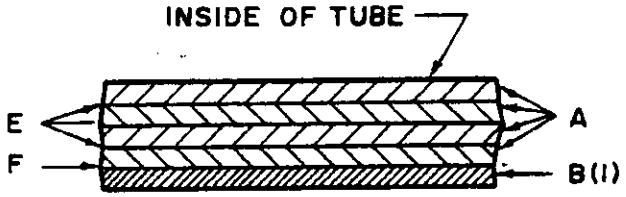
FIGURE 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:56Z
Check the source to verify that this is the current version before use.

CONSTRUCTION OF NECK TUBES

FOR

TYPE I, II AND III CONTAINERS



A - AMMUNITION CONTAINER BOARD, SPECIFICATION MIL-B-20390, AS REQUIRED TO PROVIDE SPECIFIED DIAMETER. USE ONE OR MORE LAYERS OF PAPER OF EQUAL OR BETTER GRADE WITH MINIMUM THICKNESS OF .007" TO OBTAIN SPECIFIED DIAMETER.
B - .007" WATERPROOF ASPHALT DUPLEX KRAFT WRAPPING PAPER, SPECIFICATION PPP-B-1055, Class P-1 (MIST LAP NOT LESS THAN 1/16").
E - GLUE, SPECIFICATION MMM-A-100. (NOTE G).
F - ASPHALT, SPECIFICATION MIL-A-3029.
G - ALTERNATE ADHESIVES - ADHESIVE, DEXTRIN, SPECIFICATION MIL-A-13374 (ORD), TACKIFIED POLYVINYL ALCOHOL ADHESIVE OR POLYVINYL ACETATE ADHESIVE MEETING PERFORMANCE TESTS OF SPECIFICATION MMM-A-100.
(1) ENTIRE INNER SURFACE TO BE COMPLETELY COATED WITH ASPHALT.

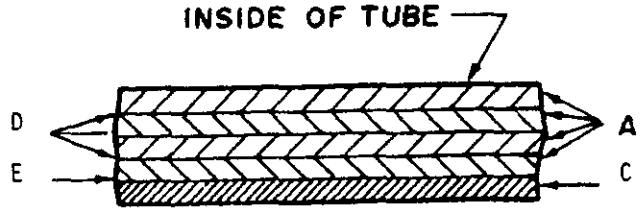
FIGURE 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:56Z
Check the source to verify that this is the current version before use.

CONSTRUCTION OF NECK TUBES

FOR

TYPE IV CONTAINERS



A - Ammunition container board, specification MIL-B-20390, as required to provide specified diameter. Use one or more layers of paper of equal or better grade with minimum thickness of .007" to obtain specified diameter.

C - 7 point lusterless black kraft. Lap shall not be less than 5/16".

D - Glue specification MMM-A-100 (see note F)

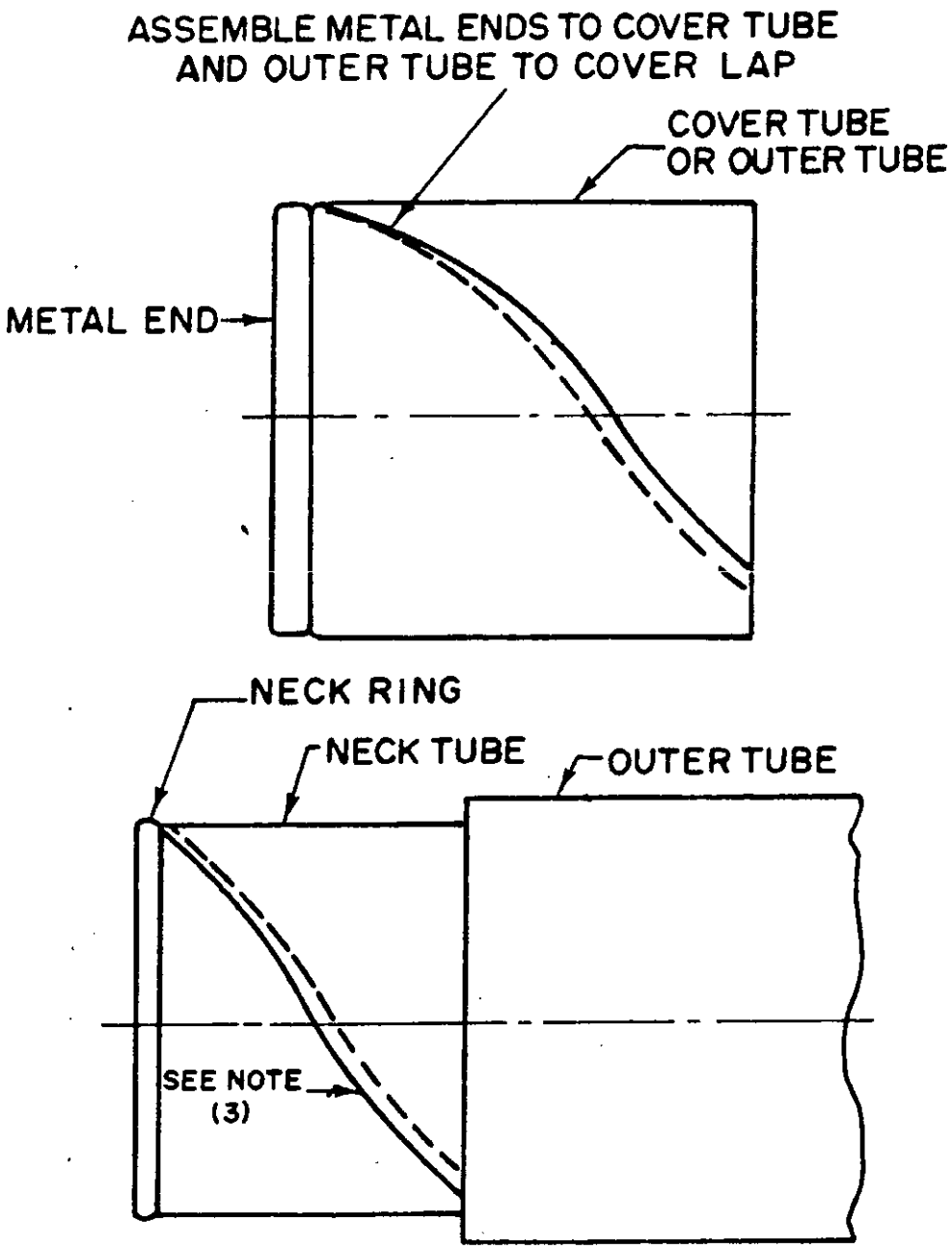
E - Polyvinyl acetate adhesive, product no. 53-1430, supplied by United Resin Products, Inc., Brooklyn, NY or equivalent.

F - Alternative Adhesive - neoprene latex base adhesive or polyvinyl acetate adhesive meeting requirements of specification MIL-A-45059.

G - Resin is applied to all the outer surfaces of the tubes.

FIGURE 4

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:56Z
Check the source to verify that this is the current version before use.

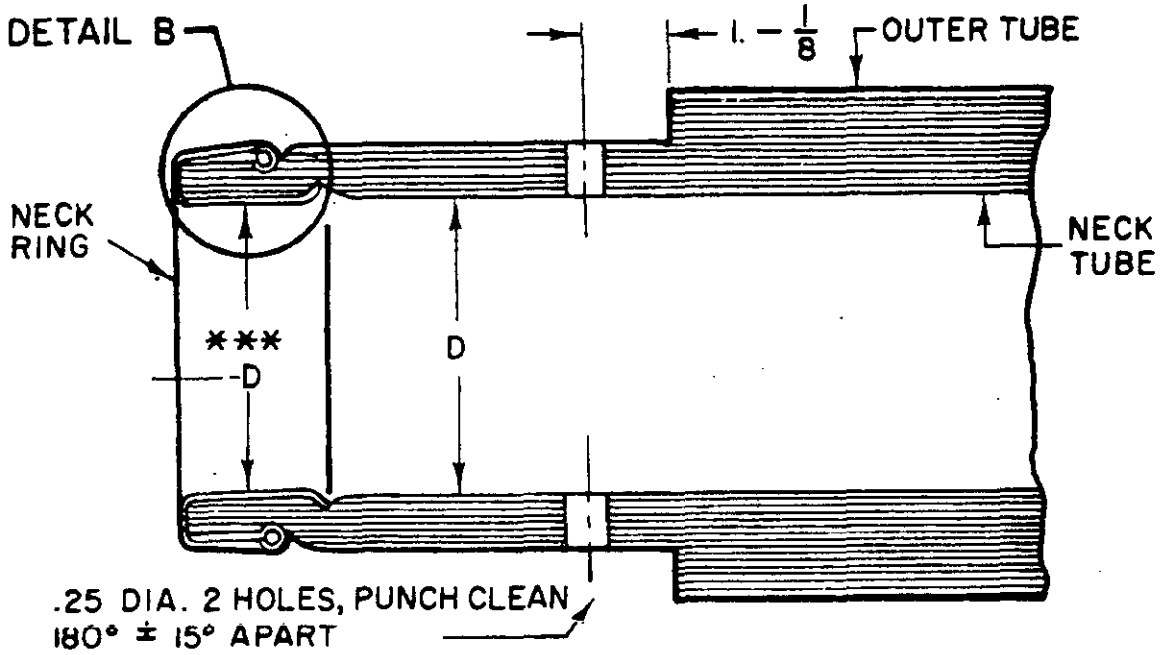


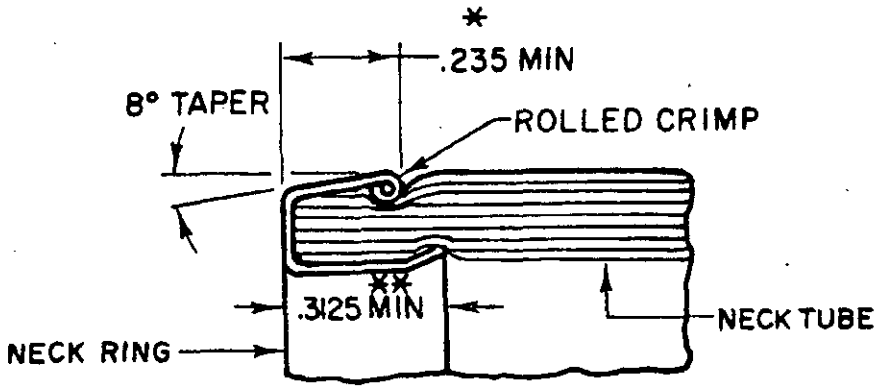
(3) OVERLAP MAY BE REVERSED WHEN NECK RING IS USED

FIGURE 5

Source: https://assist.dla.mil -- Downloaded: 202610-16T14:56Z
Check the source to verify that this is the current version before use.

NECK RING DETAILS
AND
NECK TUBE AIR RELIEF HOLES





DETAIL "B"

* .125 FOR 5.25 ID OR LESS
** .1875 FOR 5.25 ID OR LESS
*** DIAMETER = D - TWICE THICKNESS OF METAL (SEE PAR. 3.6)

FIGURE 6

SEE PARAGRAPH 3.6 FOR ADDITIONAL DETAILS

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:56Z
Check the source to verify that this is the current version before use.

POLYESTER TAPE APPLIED TO NECK TUBE

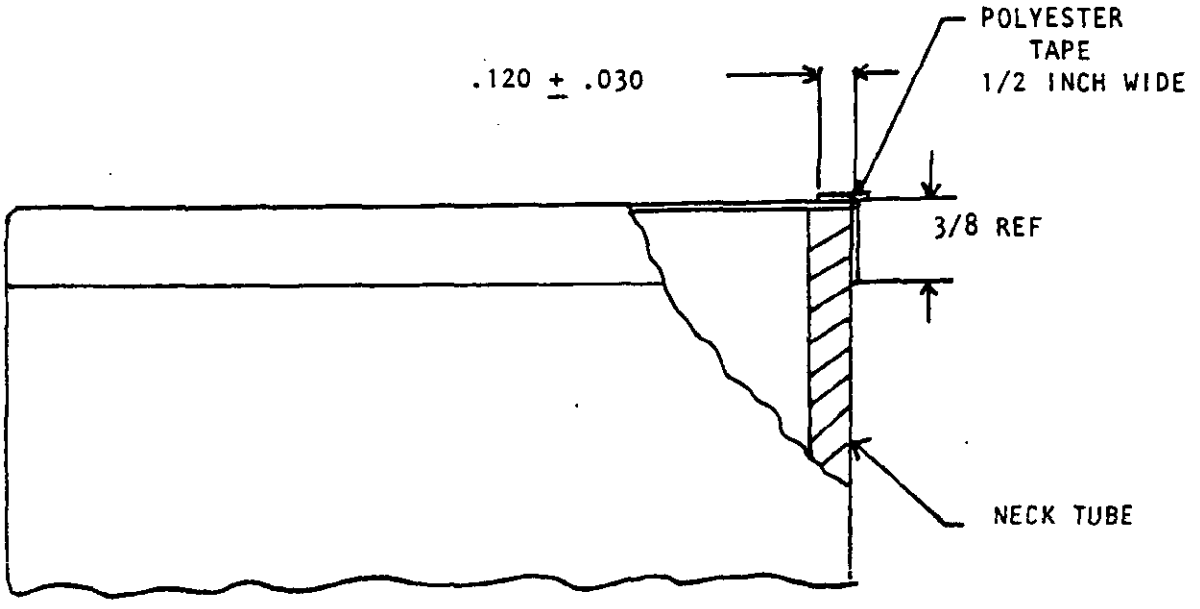
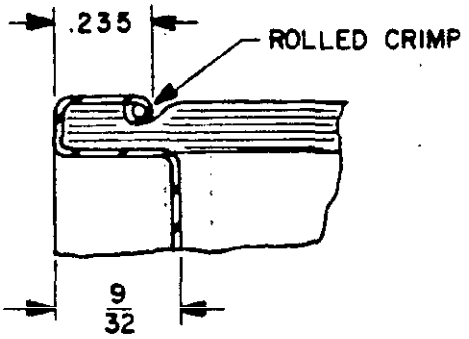


FIGURE 7

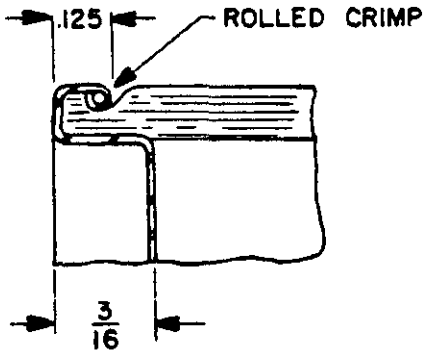
NOTES:

1. MINIMUM OVERLAP OF POLYESTER TAPE SHALL BE 1/4 INCH.
2. PEEL STRENGTH TEST SHALL BE APPLICABLE TO THE PORTION OF THE POLYESTER TAPE WHICH IS ADHERED TO THE SIDEWALL OF THE NECK TUBE ONLY.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:56Z
Check the source to verify that this is the current version before use.

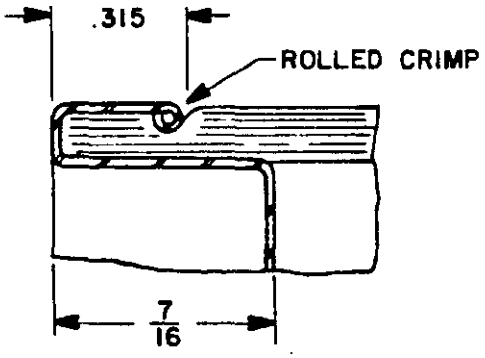


MORE THAN 3 INCHES



3 INCHES OR LESS

ALL DIMENSIONS ON SKETCHES ARE MINIMUM.



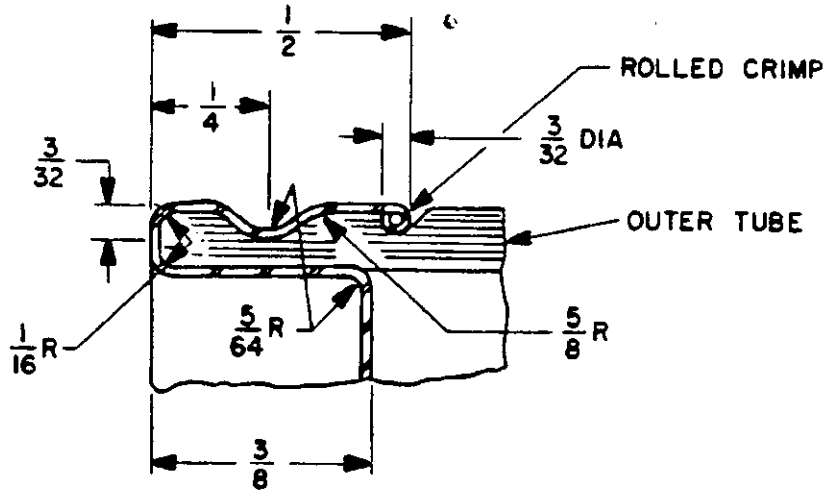
3.39 INCHES OR MORE
WITH 26 GAGE METAL ENDS.

FIGURE 8

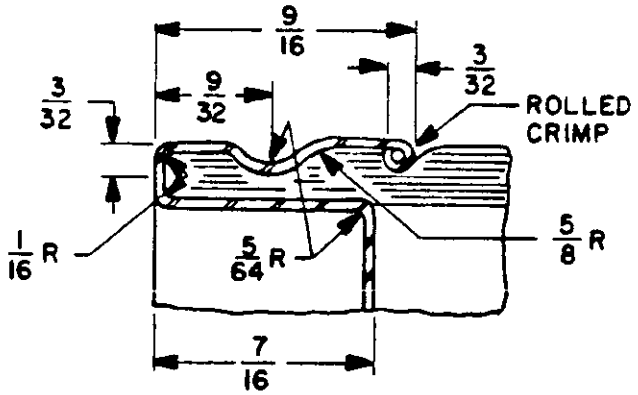
CLASS 1 METAL ENDS

METAL END CRIMP DETAILS FOR OUTER TUBE INSIDE DIAMETERS AS SHOWN

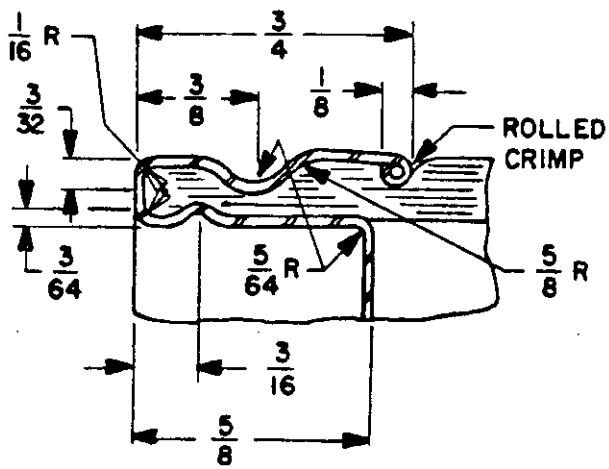
Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:56Z
Check the source to verify that this is the current version before use.



CONTAINER WITH INSIDE DIA OF 3.39 OR LESS (WITH 26 GAGE METAL ENDS).



CONTAINER WITH INSIDE DIA OF 3.40 TO 4.70 (WITH 26 GAGE METAL ENDS).



CONTAINER WITH INSIDE DIA OF 4.71 OR GREATER (WITH 26 GAGE METAL ENDS).

TOLERANCE ON RADII ± .25%.
TOLERANCE ON ALL FRACTIONAL DIMENSIONS ± 1/32.

METAL END CRIMP DETAILS FOR OUTER TUBE INSIDE DIAMETERS AS SHOWN

FIGURE 9

Source: https://assist.dla.mil - Downloaded: 2026-10-16T14:56Z
Check the source to verify that this is the current version before use.

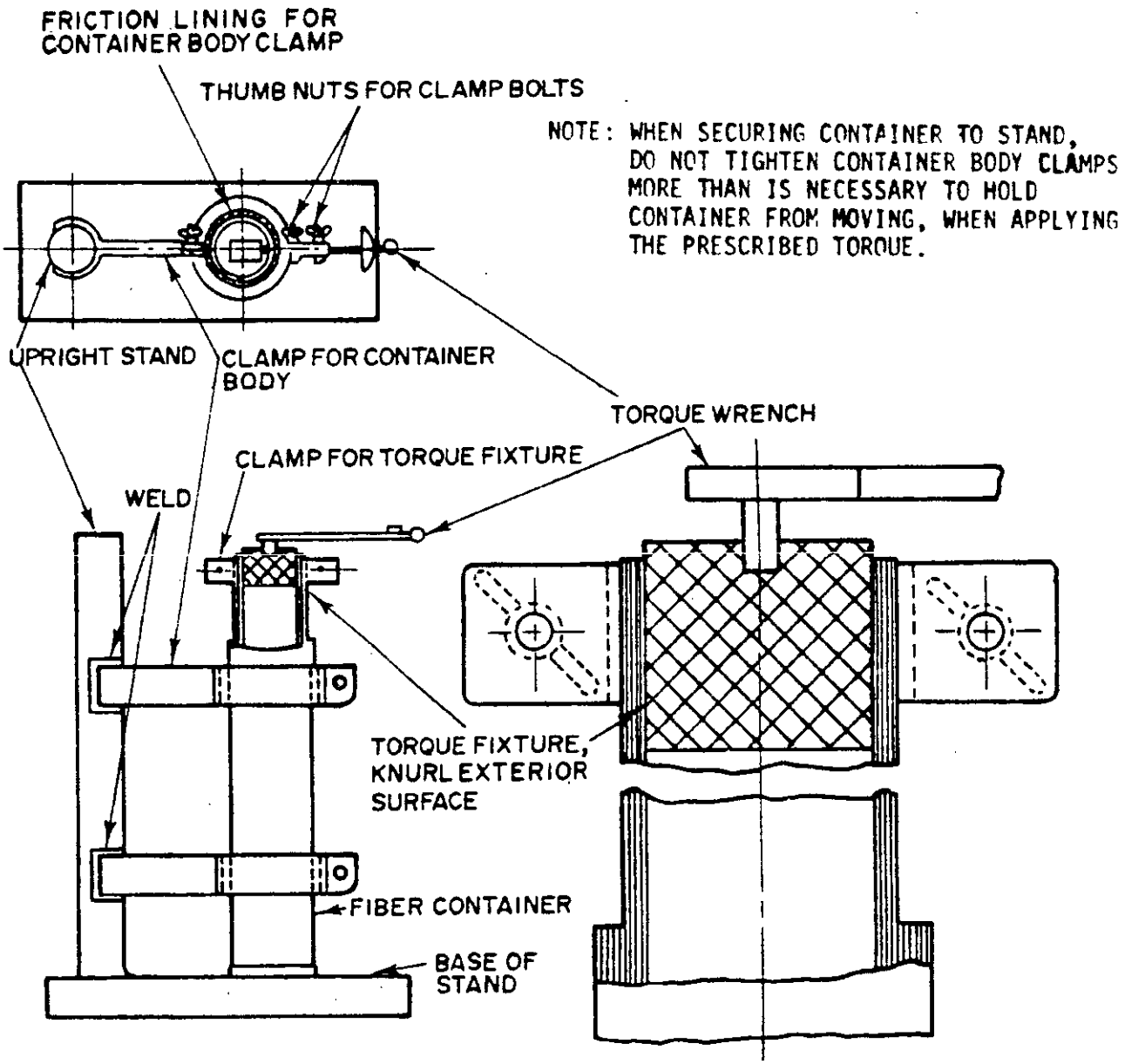
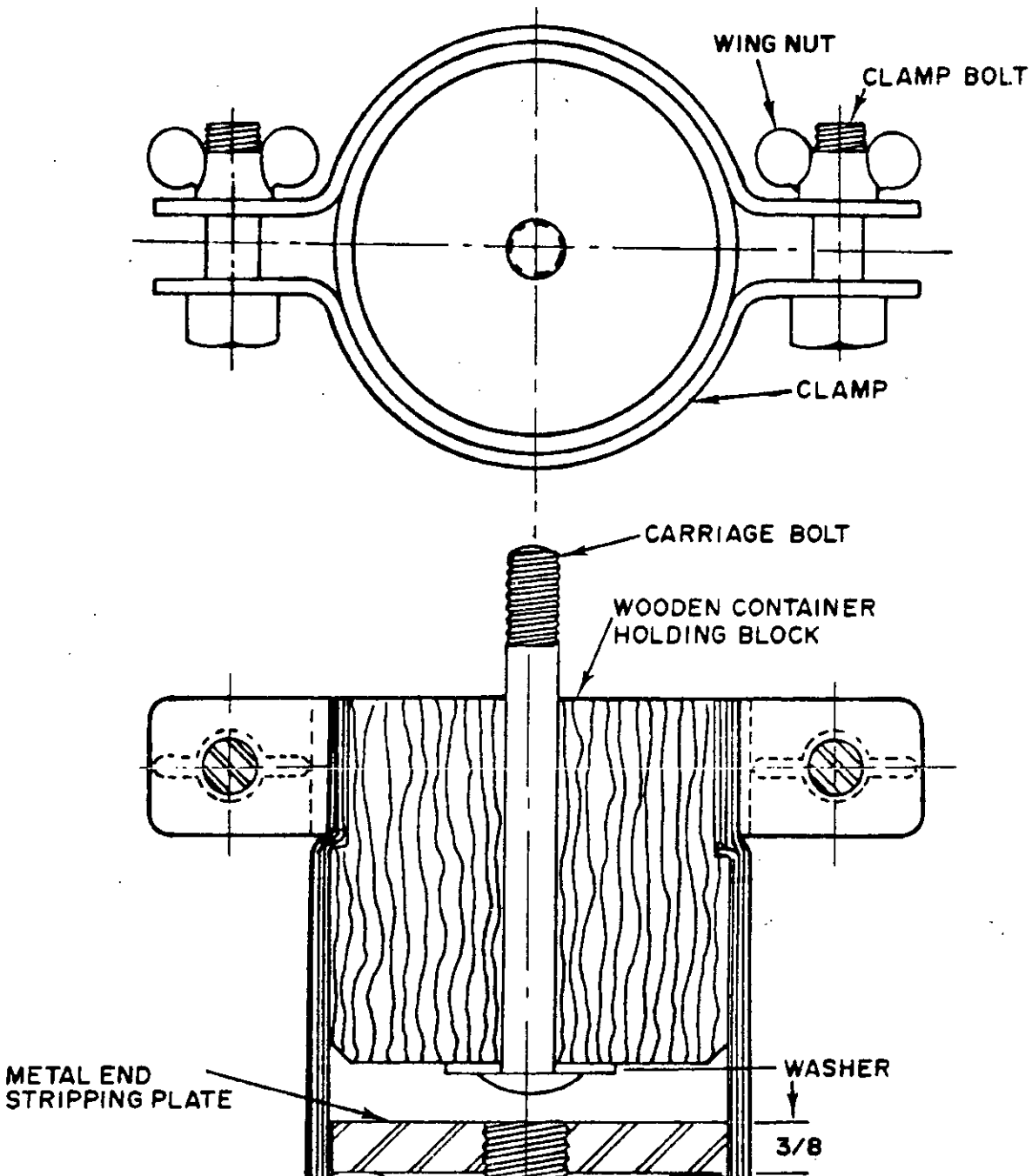


FIGURE 10

Source: https://assist.dla.mil -- Downloaded: 20260-16T14:56Z
Check the source to verify that this is the current version before use.

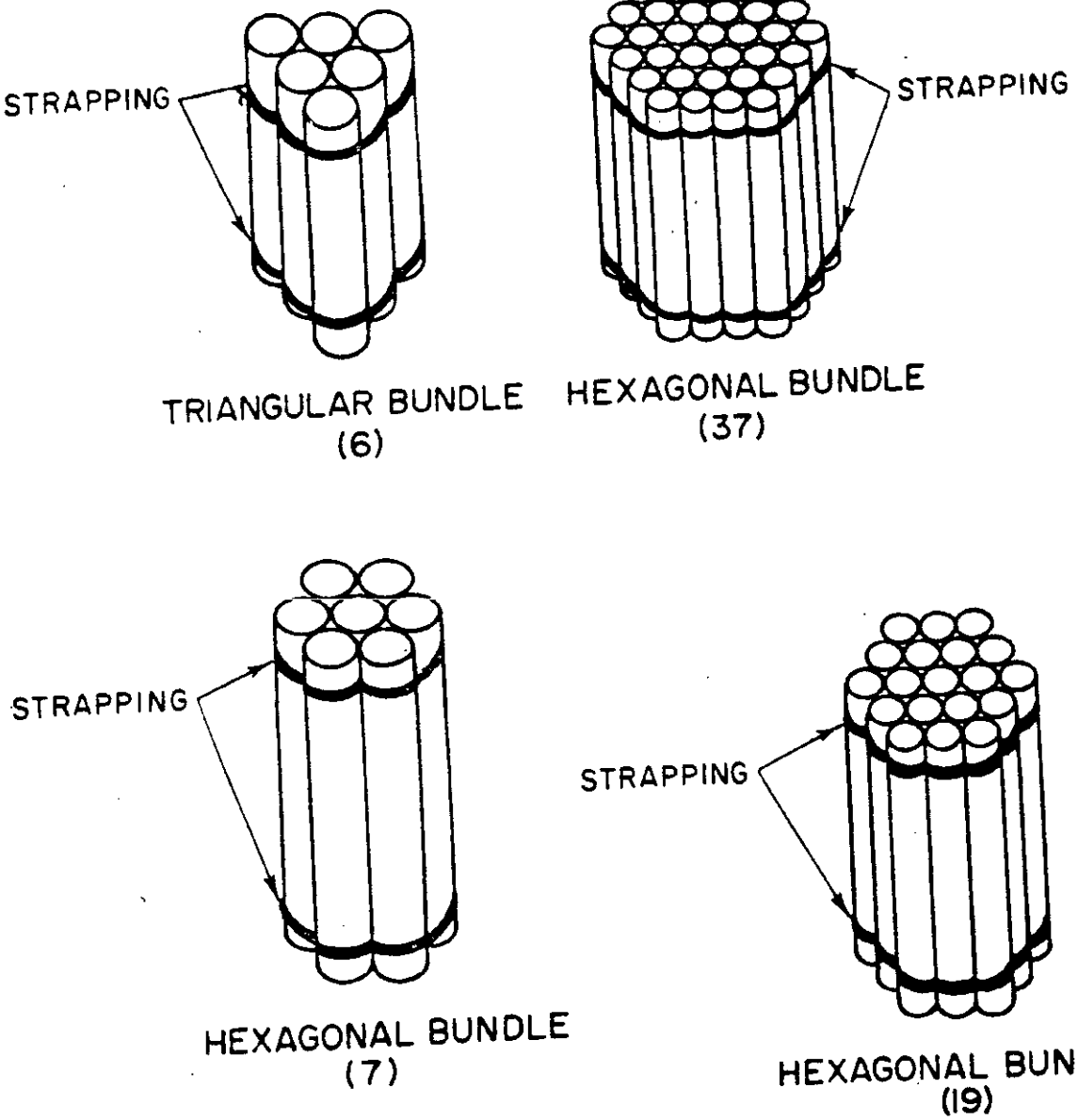


FIXTURE FOR STRIPPING METAL END FROM FIBER CONTAINERS

FIGURE 11

Source: https://assist.dla.mil -- Downloaded: 202610-16T14:56Z
Check the source to verify that this is the current version before use.

AMMUNITION CONTAINERS
TYPES OF BUNDLES



TRIANGULAR BUNDLE
(6)

HEXAGONAL BUNDLE
(37)

HEXAGONAL BUNDLE
(7)

HEXAGONAL BUNDLE
(19)

FIGURE 12

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:56Z
Check the source to verify that this is the current version before use.

Custodian:
Army-AR
Navy-OS

Review Activities:
Army-AR, SL
Navy-OS

Preparing activity:
Army-AR

(Project 8140-0715)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:56Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
"PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

COMMANDER
US ARMY ARMAMENT RSCH DEV & SNGR CENTER
ATTN: SMCAR-QA
DOVER, NJ 07801 5001

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:56Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
M L-C-2439E	CONTAINER, AMMUNITION, FIBER, SPIRALLY WOUND

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:56Z
Check the source to verify that this is the current version before use.